

Foundation (Jalapeno)

- Q1.** Solve for x : $x^2 + 4x + 4 = 0$ using the quadratic formula. What is the discriminant?
- (A) 0
(B) 4
(C) 8
(D) 12
(E) 16
- Q2.** A company's profit P is given by $P = -2x^2 + 12x - 5$. Find the discriminant to determine the nature of solutions for x when $P = 0$.
- (A) 104
(B) 112
(C) 128
(D) 144
(E) 156
- Q3.** How many solutions does $x^2 + 5x + 7 = 0$ have?
- (A) 0 real, 2 complex
(B) 1 real, 1 complex
(C) 2 real, 0 complex
(D) 1 real, 0 complex
(E) 0 real, 0 complex
- Q4.** What is the purpose of finding the discriminant in a quadratic equation?
- (A) To find the x -intercepts
(B) To find the y -intercept
(C) To determine the number of solutions
(D) To simplify the equation
(E) To graph the equation
- Q5.** A stock's value V is modeled by $V = -3x^2 + 10x + 2$. Find the discriminant when $V = 0$ to determine possible buy times.
- (A) 64
(B) 76
(C) 88
(D) 100
(E) 112
- Q6.** Solve $x^2 - 6x + 9 = 0$ and determine the nature of its solutions.
- (A) 1 real solution, 0 complex
(B) 2 real solutions, 0 complex
(C) 0 real solutions, 2 complex
(D) 2 real solutions, 2 complex
(E) 0 real solutions, 0 complex
- Q7.** The quadratic equation $x^2 - 4x - 3 = 0$ has how many real solutions?
- (A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q8.** A savings account grows according to $A = -2x^2 + 5x + 1000$ where A is the amount after x years. What is the discriminant when $A = 0$?
- (A) 25
(B) 33
(C) 41
(D) 49
(E) 57

Open Ended Questions (FRQ)

- Q9.** Solve for x : $2x^2 + 5x - 3 = 0$ using the quadratic formula and determine the nature of its solutions.
- Q10.** Given $x^2 + bx + c = 0$, find the values of b and c if the discriminant is 16 and one solution is $x = 2$. Show your work and explain your reasoning.

Chef's Tips

- Remind students to check their work for reasonableness and to consider the context of the problem
- Encourage students to use the quadratic formula to solve equations when factoring is not straightforward
- Emphasize the importance of determining the nature of solutions to understand the behavior of the quadratic function